SALES & BUSINESS PERFORMANCE ANALYTICS

Project Report

Sample Superstore Dataset

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Executive Summary

This project analyzes retail transaction data from the Sample Superstore dataset to assess sales, profitability, customer segments, product categories, regional performance, and time-based trends. The Orders table contains 9,994 transaction-line records representing 5,009 distinct orders and 793 customers across 03 January 2014 to 30 December 2017. Across the period, the business generated $2,297,200.86 in sales and $286,397.02 in profit, with an overall profit margin of 12.47%.

Technology is the strongest category by sales and profit, while Furniture has a substantially weaker profit margin. The West region leads sales and profit, whereas Central has the lowest regional margin. Sales and profit rise over the full four-year period, with 2017 recording the highest annual totals. Discount analysis shows that several high-discount bands are associated with negative profitability; this is an observational pattern and should not be interpreted as proof of causation.

The analysis was performed in Python using Pandas, NumPy, Matplotlib and Seaborn. Microsoft Power BI is being used to develop an interactive dashboard. Dashboard development is currently in progress.

1. Introduction

Retail businesses generate large volumes of transaction data. Converting these records into reliable performance measures helps decision-makers understand where revenue is generated, where profit is lost, and which customer or regional segments deserve further attention. This project applies a structured analytics workflow to the Sample Superstore dataset and communicates the results through statistical summaries, visualizations, and a planned Power BI dashboard.

1.1 Project Objectives

- Inspect and validate the source Orders, Returns, and People tables.
- Clean data types and standardize fields for analysis.
- Calculate core sales and profitability KPIs.
- Analyze performance by product category, region, customer segment, product, discount, and time.
- Develop an interactive Power BI dashboard and derive practical business recommendations.

2. Dataset Description

The supplied Excel workbook contains three sheets: Orders, Returns, and People. The Orders sheet is the primary transaction table. Returns identifies orders marked as returned, while People maps regional managers to regions.

Sheet	Records	Fields / role
Orders	9,994	Transactional sales, customer, product, shipping, discount and profit fields
Returns	296	Returned Order IDs
People	4	Regional people / manager mapping

The Orders table has 22 source columns before the return indicator is added. It covers 03 Jan 2014 through 30 Dec 2017. The analysis uses order-line records for sales and profit aggregation, and distinct Order IDs when counting orders.

3. Data Preparation and Quality Checks

- Standardized column names and text values for consistent analysis.
- Converted Order Date and Ship Date into date data types.
- Treated Postal Code as text to preserve postal-code formatting.
- Joined return information to Orders using Order ID and created an order-level Returned indicator.
- Created calendar attributes including Year, Month, Month Name, Quarter, and YearMonth for time analysis.
- Checked missing values, duplicate records, positive sales and quantity, discount bounds, and shipping-date consistency.

All 296 distinct Order IDs in the Returns sheet matched an order in the Orders table. The returned indicator is repeated across line items belonging to the same returned order; therefore, the 800 flagged transaction rows must not be interpreted as that many returned orders. There are 296 distinct returned orders.

4. KPI Analysis

The following KPIs were calculated from the cleaned Orders data. Monetary values retain the dataset's US-dollar convention.

KPI	Value	Calculation / interpretation
Total Sales	$2,297,200.86	Sum of Sales
Total Profit	$286,397.02	Sum of Profit
Total Orders	5,009	Distinct Order IDs
Total Customers	793	Distinct Customer IDs
Average Order Value	$458.61	Total Sales ÷ Distinct Orders
Profit Margin	12.47%	Total Profit ÷ Total Sales
Quantity Sold	37,873	Sum of Quantity

5. Exploratory Data Analysis

5.1 Product Category Performance

Technology generated the highest sales and profit. Office Supplies sold the highest quantity, while Furniture had the weakest profit margin. This indicates that sales volume alone does not represent financial performance.

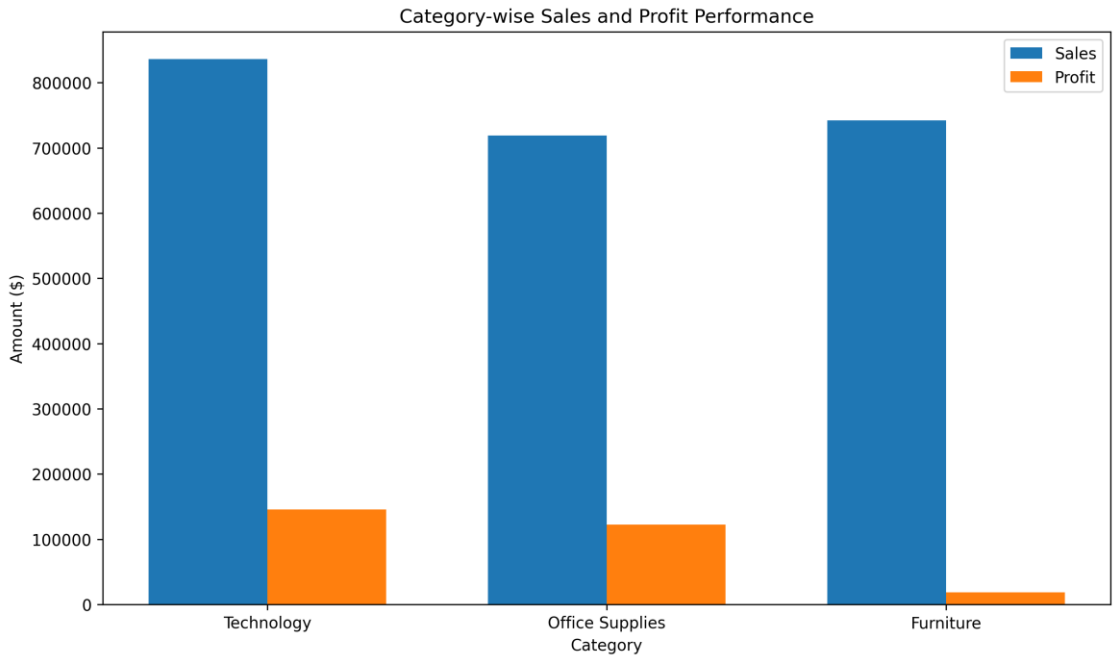


Figure 1. Sales and profit performance by product category.

Category	Sales	Profit	Quantity	Profit margin
Furniture	$741,999.80	$18,451.27	8,028	2.49%
Office Supplies	$719,047.03	$122,490.80	22,906	17.04%
Technology	$836,154.03	$145,454.95	6,939	17.40%

5.2 Regional Performance

The West region recorded the highest sales and profit. East was the second-largest sales contributor. Central generated the lowest profit margin among the four regions, suggesting an area for deeper investigation.

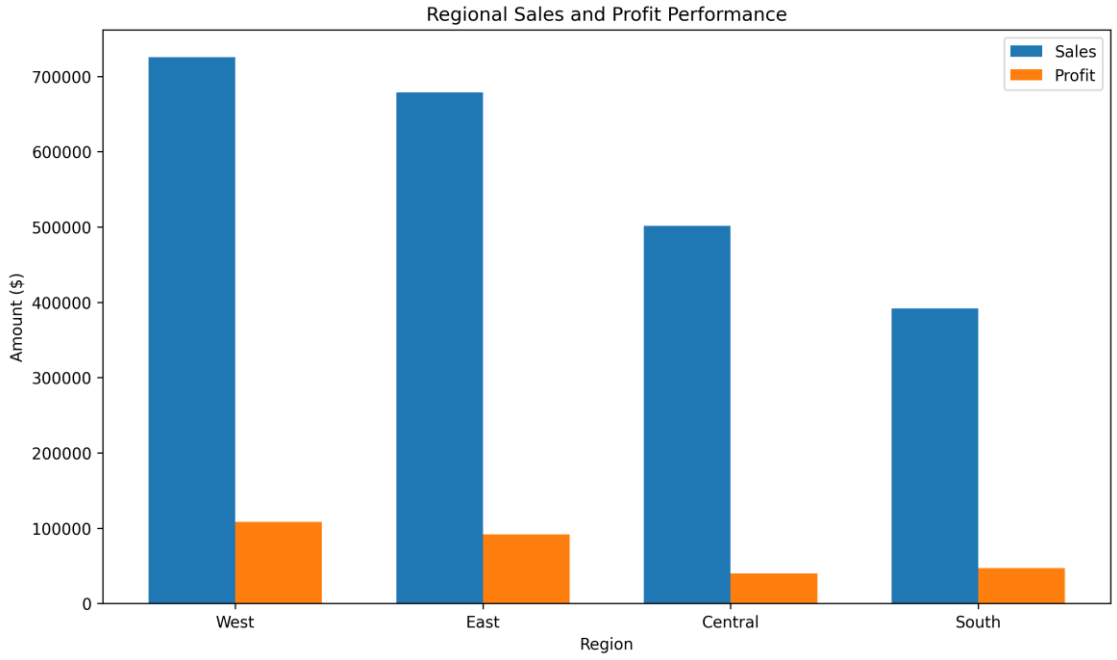


Figure 2. Sales and profit performance across regions.

Region	Sales	Profit	Orders	Profit margin
West	$725,457.82	$108,418.45	1,611	14.94%
East	$678,781.24	$91,522.78	1,401	13.48%
Central	$501,239.89	$39,706.36	1,175	7.92%
South	$391,721.91	$46,749.43	822	11.93%

5.3 Customer Segment Performance

Consumer contributed the largest sales and profit totals. Home Office recorded the highest segment profit margin and average order value. These differences can inform segment-specific sales and retention strategies.

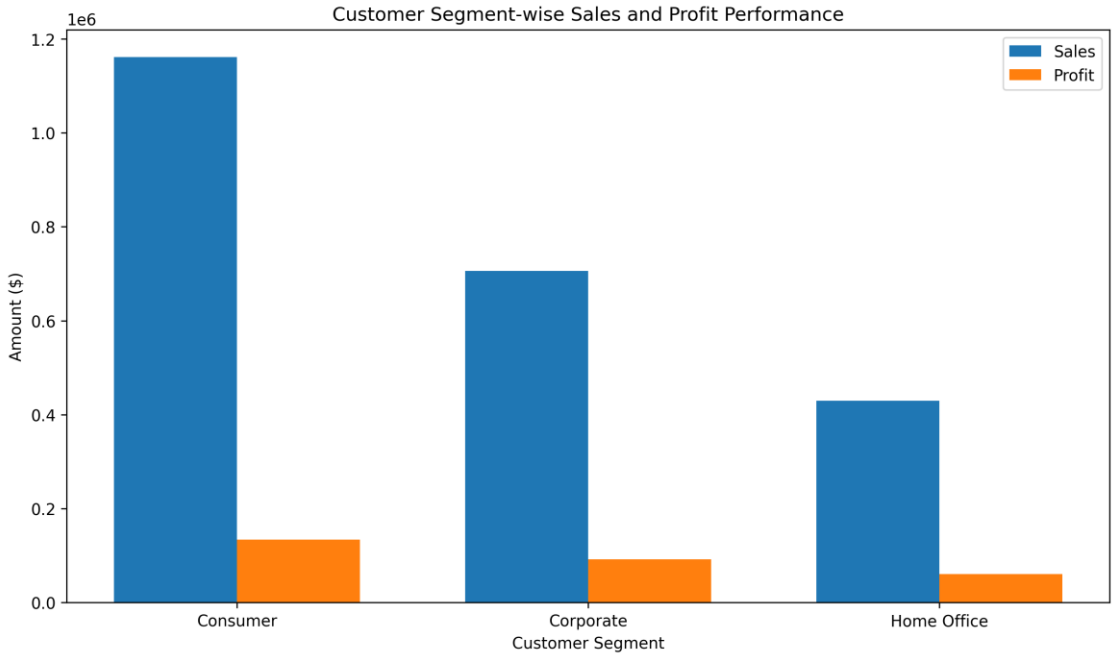


Figure 3. Sales and profit performance by customer segment.

Segment	Sales	Profit	Orders	Customers	Margin	AOV
Consumer	$1,161,401.34	$134,119.21	2,586	409	11.55%	$449.11
Corporate	$706,146.37	$91,979.13	1,514	236	13.03%	$466.41
Home Office	$429,653.15	$60,298.68	909	148	14.03%	$472.67

5.4 Time-Based Performance

Annual sales and profit increased over the analysis period. 2017 recorded the highest annual sales, profit, and order count. Monthly results show seasonality and month-to-month variation, with November 2017 recording the highest monthly sales in the dataset.

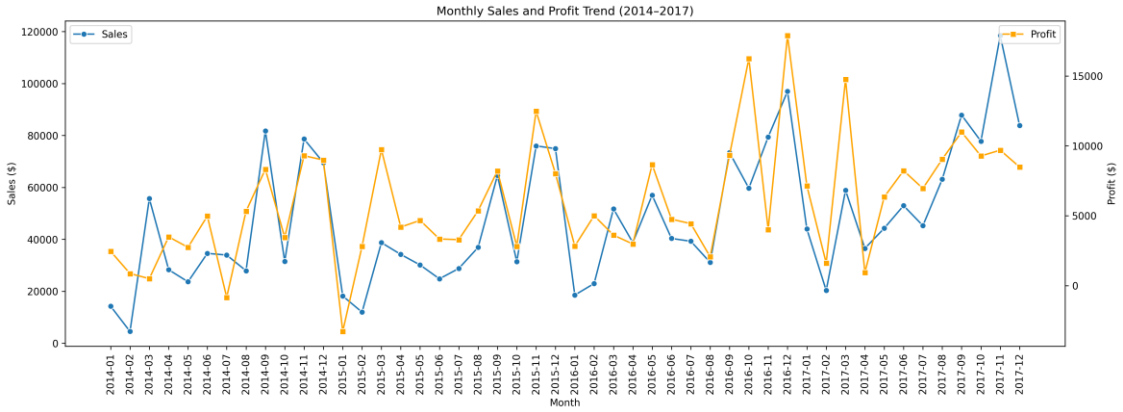


Figure 5. Monthly sales and profit trend across the analysis period.

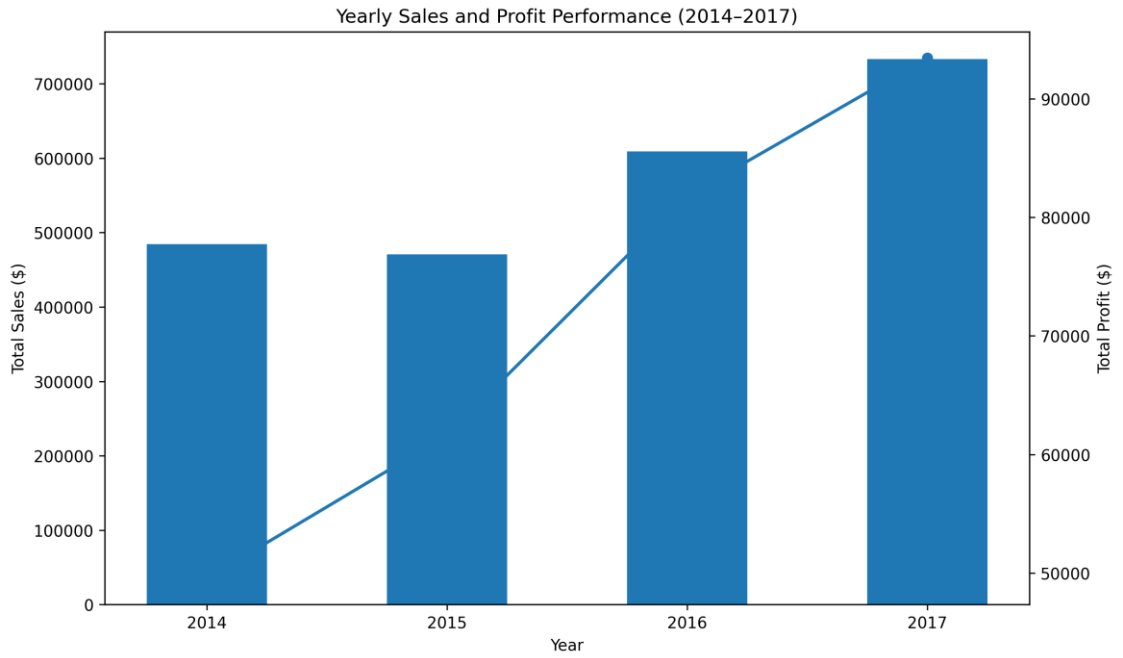


Figure 4. Annual sales and profit performance.

Year	Sales	Profit	Orders
2014	$484,247.50	$49,543.97	969
2015	$470,532.51	$61,618.60	1,038
2016	$609,205.60	$81,795.17	1,315
2017	$733,215.26	$93,439.27	1,687

5.5 Discount and Profitability

The discount analysis compares profit margin across discount levels. Several higher-discount bands show negative margins, particularly in some category and discount combinations. However, discount level may be related to product mix, transaction size, and other factors. The analysis is descriptive and does not establish that discounts alone caused the observed losses.

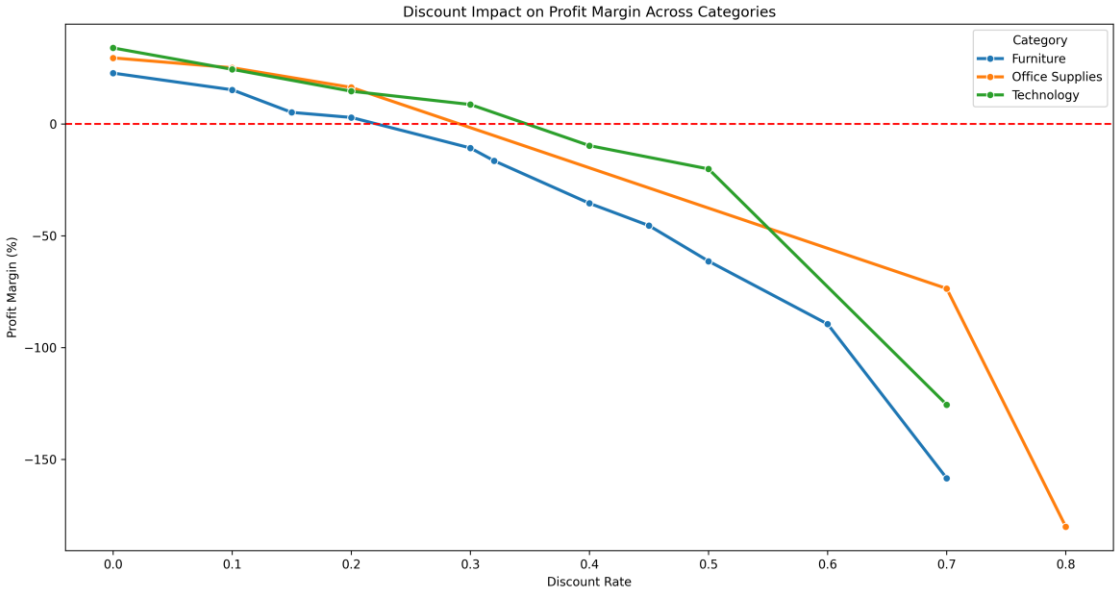


Figure 7. Discount impact across product categories.

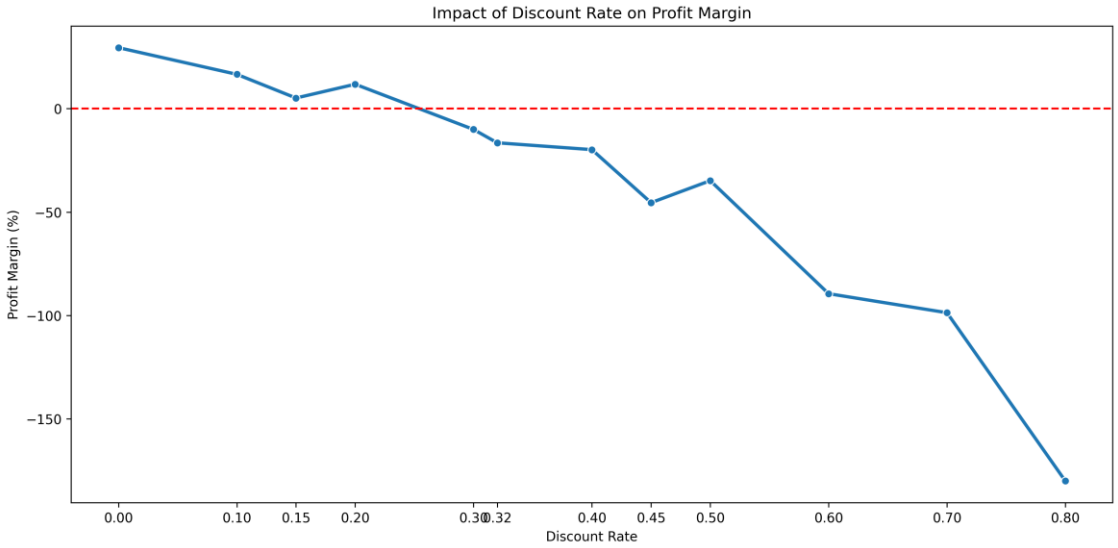


Figure 6. Relationship between discount levels and profit margin.

5.6 Product-Level Observations

The Canon imageCLASS 2200 Advanced Copier was the top product by both total sales and total profit, with sales of $61,599.82 and profit of $25,199.93. Some high-sales products recorded losses, demonstrating the importance of monitoring profitability alongside revenue.

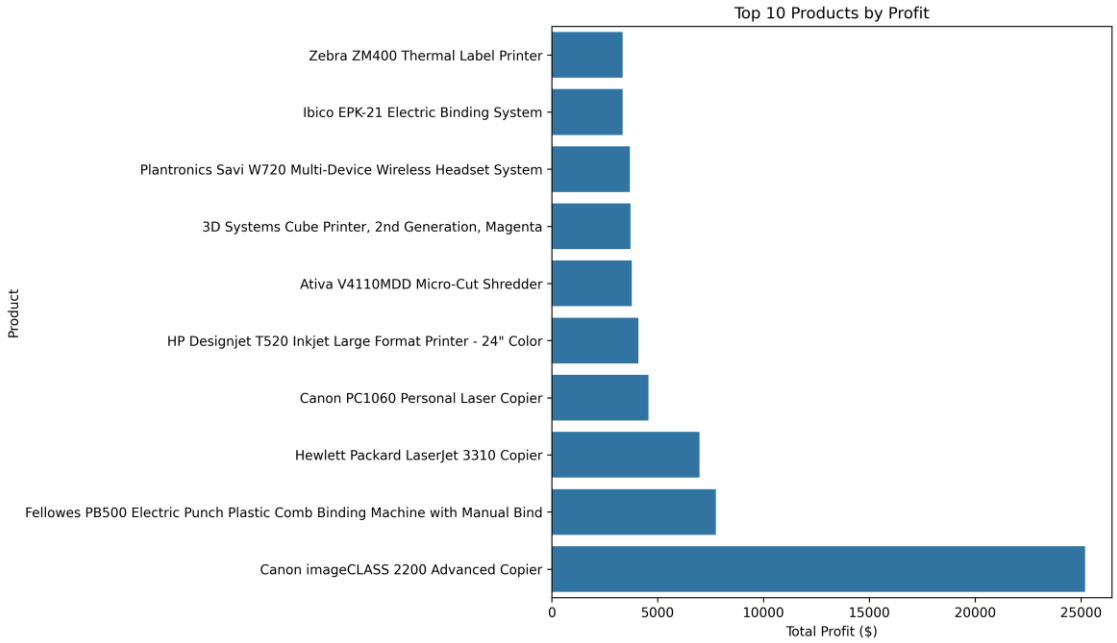


Figure 9. Top ten products ranked by profit.

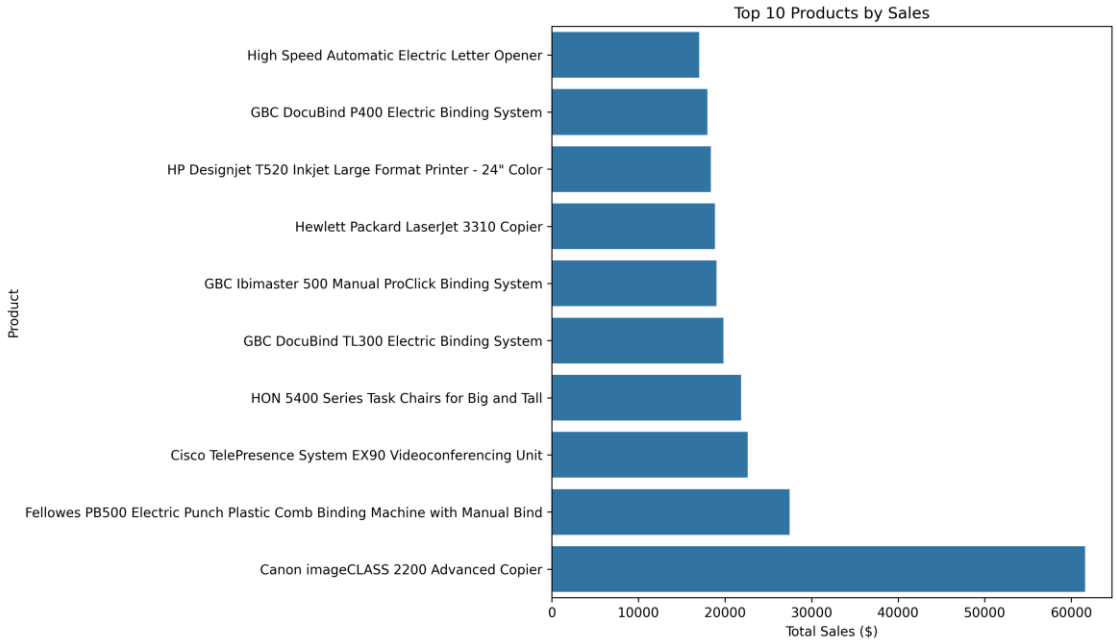


Figure 8. Top ten products ranked by sales.

High-sales product with negative profit	Sales	Profit
Cisco TelePresence System EX90 Videoconferencing Unit	$22,638.48	$-1,811.08
GBC DocuBind P400 Electric Binding System	$17,965.07	$-1,878.17
High Speed Automatic Electric Letter Opener	$17,030.31	$-262.00

6. Key Findings

- The dataset generated $2,297,200.86 in sales and $286,397.02 in profit, with a 12.47% overall profit margin.
- Technology led the categories in sales and profit; Furniture had a comparatively weak margin.
- West led regional sales and profit; Central had the lowest regional profit margin.
- Consumer generated the largest sales and profit totals, while Home Office had the highest segment margin.
- 2017 was the strongest year in total sales, profit, and order volume.
- Some high-discount combinations and high-sales products were unprofitable, warranting margin-level review.

7. Business Recommendations

- **Review Furniture economics:** Investigate product-level costs, pricing, returns, and discounting to understand the category's low margin.
- **Set discount guardrails:** Review approval thresholds for high discounts and monitor contribution margin by product and category.
- **Replicate successful Technology practices:** Examine high-performing Technology products and identify opportunities to protect and grow their contribution.
- **Investigate Central region profitability:** Break down Central's performance by state, product mix, discount, and customer segment before deciding on interventions.
- **Use segment-specific strategies:** Maintain Consumer volume while exploring higher-value opportunities in Corporate and Home Office segments.
- **Monitor both revenue and profit:** Use sales, profit, and margin together to avoid prioritizing high-revenue but loss-making products.

8. Power BI Dashboard

The interactive Power BI dashboard is currently under development. The team is working on its layout, visualizations, KPI measures, and interactive functionality. The dashboard will be reviewed and validated before the project is finalized.

9. Limitations

- **Dataset Scope and Generalizability:** The Sample Superstore dataset gives us a useful view of retail sales, but it represents only a sample of business activity. The patterns we found may not be the same for every retailer, especially those with different customers, products, or market conditions.
- **Limited Business Context:** The dataset shows what happened in each transaction, but not always why it happened. For example, it does not explain why a particular discount was offered, why some products made a loss, or what caused differences in performance across regions.
- **Observational Discount Analysis:** We observed a relationship between discount levels and profitability, but this does not mean that discounts were the only reason for a profit or loss. Other factors, such as the products sold and the details of each order, may also have influenced the results.
- **Return Data Granularity:** The returns data identifies returned orders, while the main dataset contains individual order-line records. As one order can contain several products, the same returned order may appear on multiple rows. For an accurate return count, we therefore count distinct Order IDs rather than the flagged rows.

10. Conclusion

The analysis demonstrates how transaction-level retail data can be transformed into business performance insights. The results identify Technology as the strongest product category by sales and profit, highlight Furniture and the Central region as areas for profitability review, and show growth in annual performance through 2017. The findings provide a basis for a Power BI dashboard and targeted business recommendations. Further investigation using product-, region-, and discount-level detail is recommended before implementing operational or pricing changes.